



Province of the
EASTERN CAPE
EDUCATION

Iphondo leMpuma Kapa: Isebe leMfundo
Provinsie van die Oos Kaap: Departement van Onderwys
Porafensie Ya Kapa Botjhabela: Letapha la Thuto

**NATIONAL
SENIORCERTIFICATE/
NASIONALE
SENIORSERTIFIKAAT**

GRADE/*GRAAD* 11

NOVEMBER 2024

**TECHNICAL MATHEMATICS P2/*TEGNIJSE WISKUNDE V2*
MARKING GUIDELINE/*NASIENRIGLYN***

MARKS/*PUNTE*: 150

This marking guideline consists of 18 pages./
Hierdie nasienriglyn bestaan uit 18 bladsye.

NOTE:

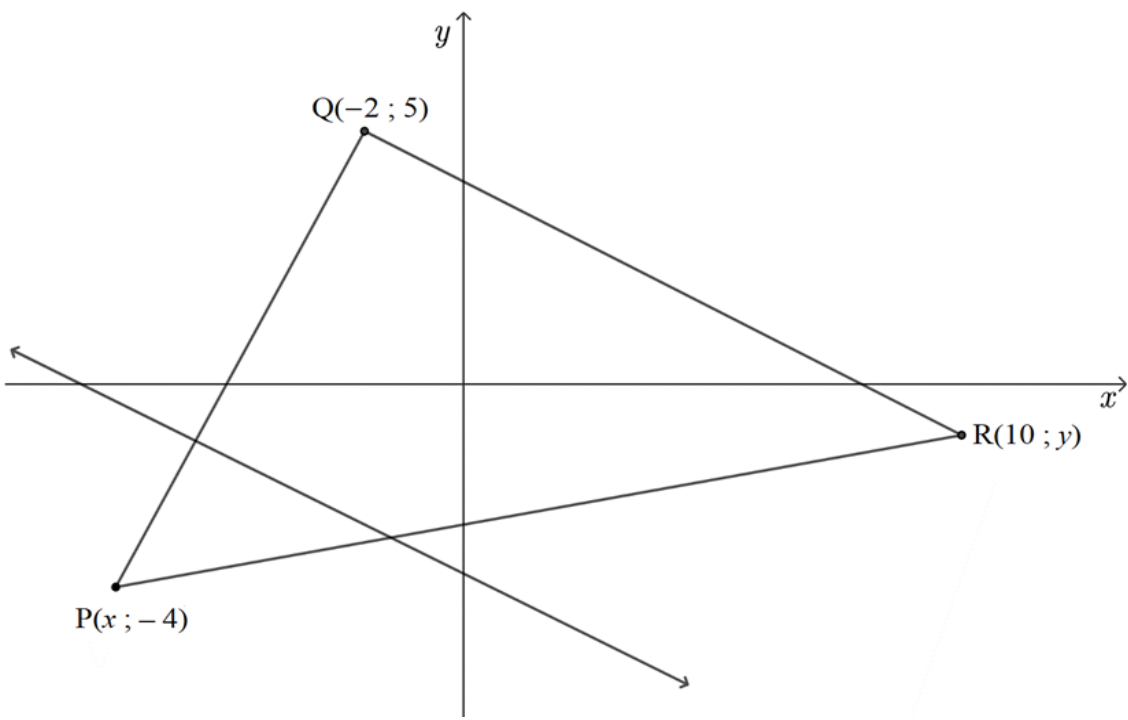
- Continuous accuracy (CA) applies only where indicated in this marking guideline.
- Assuming values/answers in order to solve a problem is unacceptable.

LET WEL:

- *Volgehoue akkuraatheid (CA) is slegs van toepassing soos aangedui in hierdie nasienriglyn.*
- *Aanvaarding van waardes/antwoorde om 'n probleem op te los, is onaanvaarbaar.*

MARKING CODES / NASIENKODES	
M	Method / Metode
A	Accuracy / Akkuraatheid
AO	Answer only / Slegs antwoord
CA	Consistent accuracy / Deurlopende akkuraatheid
F	Formula / Formule
I	Identity / Identiteit
R	Rounding / Afronding
S	Simplification / Vereenvoudiging
ST	Statement / Bewering
RE	Reason / Rede
ST RE	Statement and correct reason / Bewering en korrekte rede
SF	Substitution correctly in correct formula / Korrekte vervanging in die korrekte formule
NPU	No penalty for omitting units / Geen penalisering vir eenhede weggelaat

QUESTION/VRAAG 1

			
1.1	$2y + x + 5 = 0$ $2y = -x - 5$ $y = -\frac{1}{2}x - \frac{5}{2}$ $m_{QR} = -\frac{1}{2}$	✓ S A ✓ Ans / Ant CA	(2)
1.2	$m = \frac{y_2 - y_1}{x_2 - x_1}$ $\frac{y - 5}{10 - (-2)} = -\frac{1}{2}$ $2y - 10 = -12$ $2y = -12 + 10$ $y = -1$ $\therefore R(10; -1)$	✓ M A ✓ S A ✓ Ans / Ant CA	(3)
1.3	$M\left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$ $= M\left(\frac{-2 + 10}{2}; \frac{5 + (-1)}{2}\right)$ $= M(4; 2)$	✓ x-value / x-waarde A ✓ y-value / y-waarde A	(2)

1.4	$m_{\perp} \times m_{QR} = -1$ $m_{\perp} \times -\frac{1}{2} = -1$ $m_{\perp} = 2$ $y - y_1 = m(x - x_1)$ $y - 2 = 2(x - 4)$ $y = 2x - 8 + 2$ $y = 2x - 6$	✓ F A ✓ $m_{\perp} = 2$ A ✓ SF CA ✓ S CA ✓ Ans / Ant CA	(5)
1.5.1	Isosceles triangle / Gelykbenige driehoek	✓ Ans / Ant A	(1)
1.5.2	$QP = QR$ $\therefore QP^2 = QR^2$ $QP^2 = (x + 2)^2 + (-4 - 5)^2$ $QP^2 = x^2 + 4x + 85$ $QR^2 = (10 + 2)^2 + (-1 - 5)^2$ $QR^2 = 180$ $x^2 + 4x + 85 = 180$ $x^2 + 4x - 95 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $x = \frac{-4 \pm \sqrt{4^2 - (4 \times 1 \times -95)}}{2(1)}$ $x = -11,95 \quad \text{or / of } x = 7,95$ $\therefore x = -12$	✓ QP^2 A ✓ QR^2 A ✓ $QP^2 = QR^2$ CA ✓ Std Form / Std Vorm CA ✓ SF CA ✓ $x = -12$ CA	(6)
1.5.3	$PR = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $PR = \sqrt{(-12 - 10)^2 + (-4 - (-1))^2}$ $PR = \sqrt{493}$ $PR = 22 \text{ units / eenhede}$	✓ SF CA ✓ S CA ✓ PR CA	(3)
			[22]

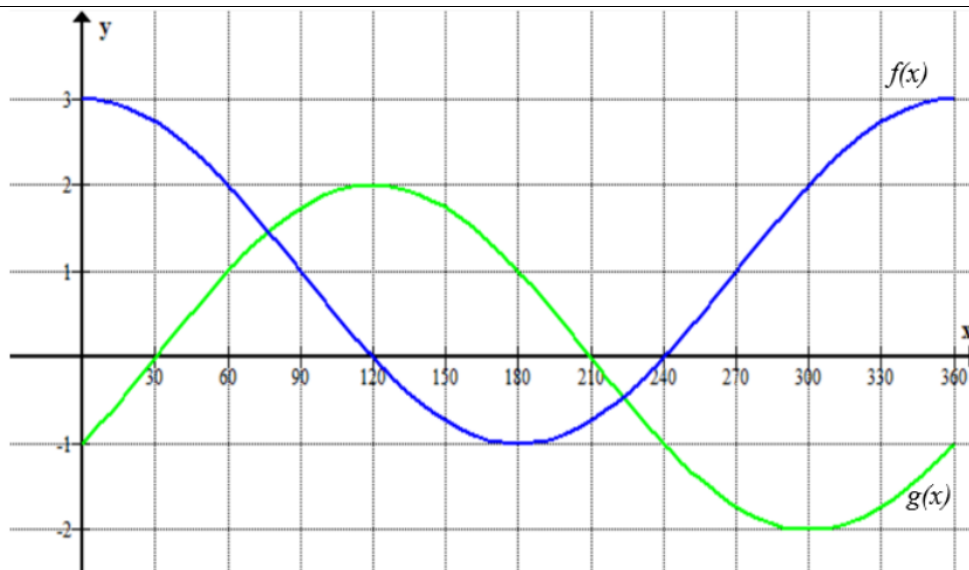
QUESTION/VRAAG 2

2.1.1	$\sin(31,24^\circ) + \cos(66,27^\circ)$ $= 0,92$	✓ Subst ✓ Ans / Ant	A CA	(2)
2.1.2	$\frac{\cot Y + \tan^2 Z}{\sec Y \cdot \operatorname{cosec} Z}$ $= \frac{\cot(31,24^\circ) + \tan^2(66,27^\circ)}{\sec(31,24^\circ) \cdot \operatorname{cosec}(66,27^\circ)}$ $= \frac{1}{\tan(31,24^\circ)} + \tan^2(66,27^\circ)$ $= \frac{1}{\cos(31,24^\circ)} \cdot \frac{1}{\sin(66,27^\circ)}$ $= 5,34$	✓ Subst ✓ S ✓ Ans / Ant	A CA CA	(3)
2.2.1	$\sin \beta = \frac{y}{r} = \frac{4}{5}$	✓ Ans / Ant	A	(1)
2.2.2	$x^2 + y^2 = r^2$ $p^2 + (4)^2 = (5)^2$ $p^2 = 25 - 16$ $p = \pm\sqrt{9}$ $\therefore p = -3$	✓ Pyth ✓ Subst ✓ S ✓ $p = -3$	A A CA CA	(4)
2.2.3	$\frac{\sec^2 \beta}{\cot \beta} = \frac{\left(\frac{5}{-3}\right)^2}{\frac{-3}{4}}$ $= \frac{25}{9} \times \frac{4}{-3}$ $= -\frac{100}{27}$	✓ $\left(\frac{5}{-3}\right)^2$ ✓ $\frac{-3}{4}$ ✓ Ans / Ant	A CA CA	(3)

2.3	$3 \tan x + 4 = 3$ $\tan x = \frac{3-4}{3}$ $\text{Ref } \angle = \tan^{-1}\left(\frac{1}{3}\right) = 18,43^\circ$ 2^{nd} Quadrant / Kwadrant $x = 180^\circ - 18,43^\circ$ $x = 161,57^\circ$ 4^{th} Quadrant / Kwadrant $x = 360^\circ - 18,43^\circ$ $x = 341,57^\circ$	$\checkmark$ S A $\checkmark \tan^{-1}\left(-\frac{1}{3}\right)$ CA $\checkmark$ Ref / Verw < CA $\checkmark x = 161,57^\circ$ CA $\checkmark x = 341,57^\circ$ CA	(5)
			[18]

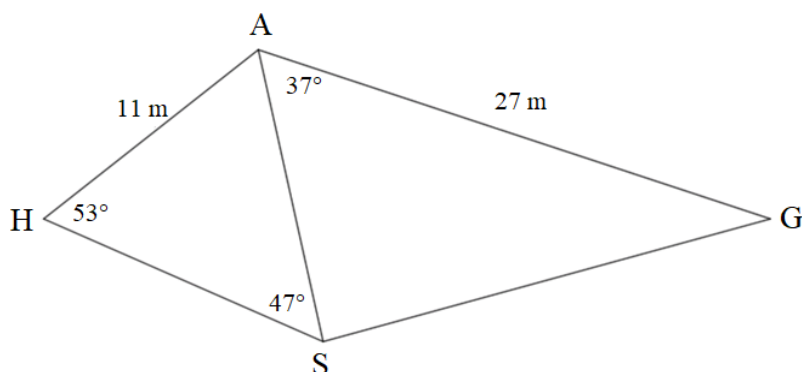
QUESTION/VRAAG 3				
3.1.1	$\cot^2 \theta$	✓ Ans / Ant	A	(1)
3.1.2	1	✓ Ans / Ant	A	(1)
3.2	$\frac{\tan^2(360^\circ+x) \cdot \sin(180^\circ+x) \cdot \cos(-x)}{\cos(180^\circ+x) \cdot \sin(360^\circ-x) \cdot \tan 315^\circ}$ $= \frac{\tan^2 x \cdot -\sin x \cdot \cos x}{(-\cos x) \cdot (-\sin x) \cdot (-1)}$ $= \tan^2 x$	✓ $\tan^2 x$ A ✓ $-\sin x$ A ✓ $\cos x$ A ✓ $-\cos x$ A ✓ $-\sin x$ A ✓ -1 CA ✓ Ans / Ant CA		(7)
3.3	$LHS = \frac{\sin x}{1 - \cos x} - \frac{\sin x}{1 + \cos x}$ $= \frac{\sin x(1 + \cos x) - \sin x(1 - \cos x)}{1 - \cos^2 x}$ $= \frac{\sin x + \sin x \cos x - \sin x + \sin x \cos x}{1 - \cos^2 x}$ $= \frac{2 \sin x \cos x}{\sin^2 x}$ $= \frac{2 \cos x}{\sin x}$ $= 2 \cot x$ $= RHS$	✓ S A ✓ $1 - \cos^2 x$ A ✓ $2 \sin x \cos x$ A ✓ $\sin^2 x$ A ✓ $\frac{2 \cos x}{\sin x}$ CA		(5)
		[14]		

QUESTION/VRAAG 4



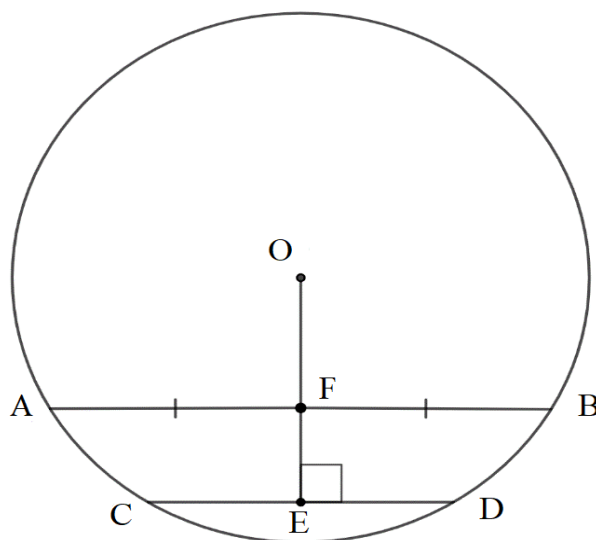
4.1	$a = 2$ $b = 2$ $q = 1$	$\checkmark a$ A $\checkmark b$ A $\checkmark q$ A	(3)
4.2	360°	$\checkmark$ Ans / Ant A	(1)
4.3	Amplitude of $g = 2$	$\checkmark$ Ans / Ant A	(1)
4.4	$x = 0^\circ$ $x = 300^\circ$ $x = 360^\circ$	$\checkmark 0^\circ$ A $\checkmark 300^\circ$ A $\checkmark 360^\circ$ A	(3)
			[8]

QUESTION/VRAAG 5



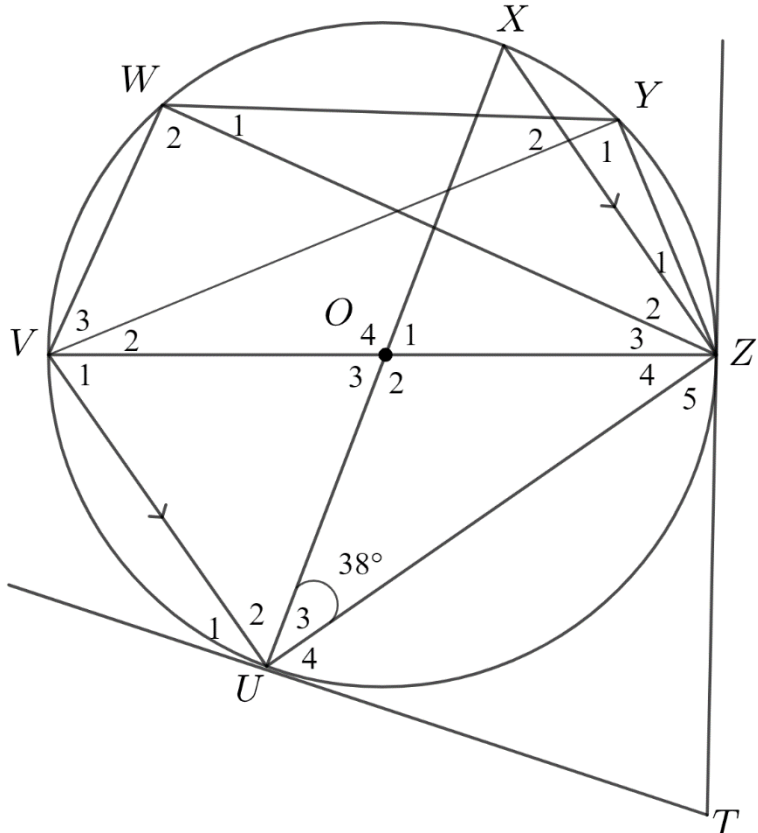
5.1	Kite	✓ Ans / Ant	A	(1)
5.2	$\frac{AS}{\sin \hat{H}} = \frac{HA}{\sin \hat{S}}$ $AS = \frac{HA \times \sin \hat{H}}{\sin \hat{S}}$ $AS = \frac{11 \times \sin 53^\circ}{\sin 47^\circ}$ $AS = 12,01 \text{ m}$	✓ Sine Rule / Reël ✓ SF ✓ Ans / Ant	A A CA	 (3)
5.3	$SG^2 = AS^2 + AG^2 - 2 \cdot AS \cdot AG \cdot \cos \hat{A}$ $SG^2 = (12,01)^2 + (27)^2 - 2(12,01)(27) \cos 37^\circ$ $SG = \sqrt{355,293...}$ $SG = 18,85 \text{ m}$	✓ Cosine Rule / Reël ✓ SF ✓ S ✓ Ans / Ant	A CA CA CA	 (4)
5.4	$\text{Area } \Delta = \frac{1}{2} \times AS \times AG \times \sin \hat{A}$ $= \frac{1}{2} \times 12,01 \times 27 \times \sin 37^\circ$ $= 97,58 \text{ m}^2$	✓ F ✓ SF ✓ Ans / Ant	A CA CA	 (3)
				[11]

QUESTION/VRAAG 6



6.1.1	$OE - 10 = OF$ $OE = x + 10$	✓ ST ✓ OE	A A	(2)
6.1.2	$OC^2 = CE^2 + OE^2$ Pythagoras Theorem $OC^2 = (24)^2 + (x + 10)^2$ $OC = \sqrt{576 + x^2 + 20x + 100}$ $OC = \sqrt{x^2 + 20x + 676}$ OR / OF $OC = OA$ Radii $OA^2 = AF^2 + OF^2$ $\therefore OC^2 = (40)^2 + x^2$ $\therefore OC = \sqrt{1600 + x^2}$	✓ ST/RE ✓ S ✓ OC	A CA CA	(3)

6.2	$OC = OA$ Radii $OC^2 = OA^2$ $x^2 + 20x + 676 = x^2 + 1600$ $20x = 1600 - 676$ $20x = 924$ $x = \frac{924}{20}$ $x = 46,2 \text{ m}$ OR / OF $OC = OA$ Radii $OC^2 = OA^2$ $x^2 + 20x + 676 = x^2 + 1600$ $20x = 1600 - 676$ $20x = 924$ $x = \frac{924}{20}$ $x = 46,2 \text{ m}$	$\checkmark$ ST $\checkmark$ RE A $\checkmark x^2 + 1600$ CA $\checkmark$ S CA $\checkmark$ Value of / <i>Waarde</i> <i>van x</i> CA OR / OF $\checkmark$ ST $\checkmark$ RE A $\checkmark x^2 + 20x + 676$ CA $\checkmark$ S CA $\checkmark$ Value of / <i>Waarde</i> <i>van x</i> CA	(5)
			[10]

QUESTION/VRAAG 7				
7.1	Diameter / <i>Middel lyn</i>	✓ Ans / <i>Ant</i>	A	(1)
7.2				
7.2.1	$\widehat{UZX} = 90^\circ$ ($\angle$ in a semi-circle / $\angle$ in <i>halwe sirkel</i>) $\widehat{VUZ} = 90^\circ$ ($\angle$ in a semi-circle / $\angle$ in <i>halwe sirkel</i>) $\widehat{OUT} = 90^\circ$ (Tan $\perp$ Rad) $\widehat{OZT} = 90^\circ$ (Tan $\perp$ Rad)	✓ ST / RE ✓ ST ✓ ST / RE ✓ ST	A A A A	(4)
7.2.2	$\widehat{V}_1 = \widehat{U}_4$ (tanchord / <i>raaklynkoord</i>) $\widehat{U}_4 = 90^\circ - 38^\circ$ (Tan $\perp$ Rad) $\therefore \widehat{V}_1 = 52^\circ$ $\widehat{X} = \widehat{V}_1 = 52^\circ$ ($\angle$'s in the same seg. / $\angle$ 'e in <i>dieselfde seg.</i>) $\widehat{U}_2 = \widehat{V}_1 = 52^\circ$ ($\angle$'s opp = sides / $\angle$ 'e <i>teenoor</i> = <i>sye</i>) $\widehat{Z}_2 + \widehat{Z}_3 = \widehat{U}_2 = 52^\circ$ ($\angle$'s in the same seg. / $\angle$ 'e in <i>dieselfde seg.</i>)	✓ ST ✓ RE ✓ $\widehat{V}_1$ ✓ ST ✓ RE ✓ ST ✓ RE ✓ ST	A A A A A	(8)
7.3.1	Tangents from same point / <i>Raaklyne vanuit dieselfde punt</i>	✓ Ans / <i>Ant</i>	A	(1)
7.3.2	$\widehat{U}_4 = \widehat{Z}_5 = 52^\circ$ ($\angle$'s opp = sides / $\angle$ 'e <i>teenoor</i> = <i>sye</i>) $T = 180^\circ - (2 \times 52^\circ)$ $T = 76^\circ$	✓ ST ✓ $\widehat{T}$	A CA	(2)

7.4	In ΔUOZ: $\hat{U}_3 = \hat{Z}_4$ ($\angle$'s opp = sides / $\angle$'e teenoor = sye) $\hat{X} = \hat{V}_1$ ($\angle$'s in the same seg. / $\angle$'e in dieselfde seg.) $\hat{Z}_{2+3+4} = \hat{U}_{2+3} = 90^\circ$ ($\angle$ in semi-circle / $\angle$ in halwe sirkel) $\Delta UXZ \equiv \Delta ZVU$ ($\lll$)	✓ ST / RE A ✓ ST A ✓ ST A ✓ RE A	(4)
7.5	$\hat{U}_1 + \hat{U}_2 = 90^\circ$ (rad $\perp$ tan) $\hat{Z}_4 + \hat{Z}_5 = 90^\circ$ (rad $\perp$ tan) $\therefore \hat{U}_1 + \hat{U}_2 = \hat{Z}_4 + \hat{Z}_5 = 90^\circ$ $\therefore$ OUTZ is a cq / is 'n kvh (ext $\angle$ of quad = opp int $\angle$ / buite $\angle$ van vh = teenoorst. binne $\angle$) OR / OF $\hat{U}_3 + \hat{U}_4 = 90^\circ$ (rad $\perp$ tan) $\hat{Z}_4 + \hat{Z}_5 = 90^\circ$ (rad $\perp$ tan) $\therefore \hat{U}_3 + \hat{U}_4 = \hat{Z}_4 + \hat{Z}_5 = 180^\circ$ $\therefore$ OUTZ is a cq / is 'n kvh (opp int $\angle$'s suppl / teenoorst. binne $\angle$'e suppl.) OR / OF OU = OZ (radii) $\hat{U}_3 = \hat{Z}_4 = 38^\circ$ ($\angle$'s opp = sides / $\angle$'e teenoor = sye) $\hat{O}_2 = 104^\circ$ (Int. $\angle$'s / binne $\angle$'e Δ) $\therefore \hat{O}_2 + \hat{T} = 180^\circ$ $\therefore$ OUTZ is a cq / is 'n kvh (opp int $\angle$'s suppl / teenoorst. binne $\angle$'e suppl.)	✓ ST / RE A ✓ ST A ✓ ST A ✓ RE A OR / OF ✓ ST / RE A ✓ ST A ✓ ST A ✓ RE A OR / OF ✓ ST / RE A ✓ ST A ✓ ST A ✓ RE A	(4)
			[24]

QUESTION/VRAAG 8

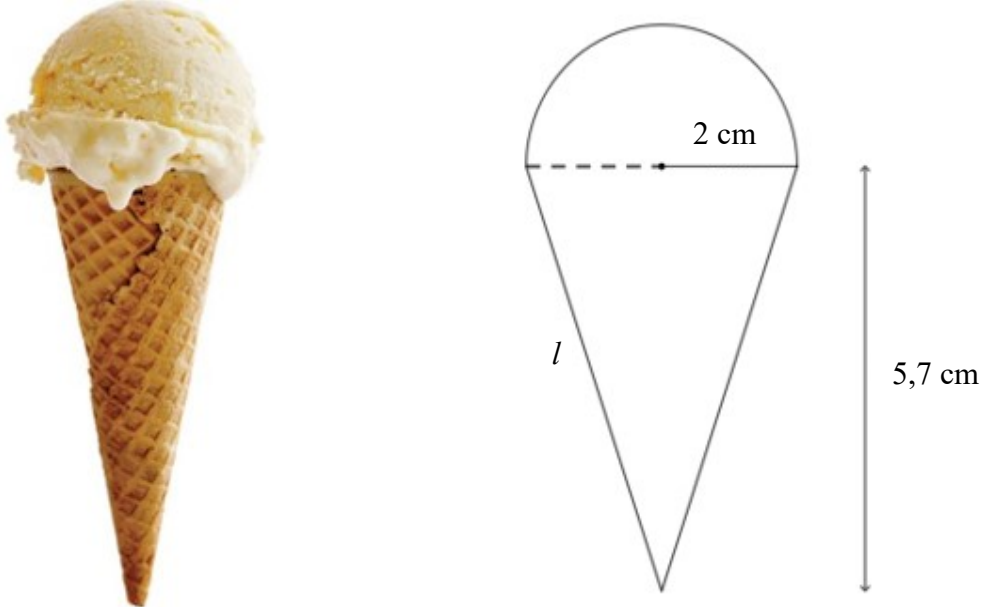
8.1	Equal / <i>Gelyk</i>	✓ Ans / <i>Ant</i> A	(1)
8.2.1	$\hat{O}_2 = 140^\circ$ ($\angle$'s on str. line / $\angle$ 'e op reguitlyn) $\hat{D}_2 = \frac{140^\circ}{2} = 70^\circ$ ($\angle$ at centre = $2 \times \angle$ at circumf. / <i>midpt</i> $\angle = 2 \times \angle$ by <i>omtrek</i>)	✓ $\hat{O}_2$ A ✓ $\hat{D}_2$ A ✓ RE A	(3)
8.2.2	$\hat{A}_1 = 70^\circ$ ($\angle$'s on str. line / $\angle$ 'e op reguitlyn OR / OF ext. $\angle$ of cq / <i>buite</i> $\angle$ van <i>kvh</i>)	✓ST✓RE A	(2)
8.2.3	$\hat{D}_1 + \hat{D}_2 = 90^\circ$ ($\angle$ in semi-circle / $\angle$ in <i>halwe sirkel</i>) $\therefore \hat{D}_1 = 90^\circ - 70^\circ$ $\therefore \hat{D}_1 = 20^\circ$	✓ST✓RE A ✓ST CA	(3)
			[9]

QUESTION/VRAAG 9

9.1	$271,314^\circ = 217^\circ + (0,314 \times 60)$ $= 217^\circ + 18' + (0,84 \times 60)$ $= 217^\circ 18' 50''$	✓ Conv. / Herlei A ✓ Ans / Ant A	(2)
9.2.1	$s = r\theta$ $\therefore \theta = \frac{s}{r}$ $\theta = \frac{15}{\left(\frac{25}{2}\right)}$ $\theta = 1,2 \text{ rad}$	✓ F A ✓ SF A ✓ radius A ✓ θ CA NPR, NPU	(4)
9.2.2	Area of sector / Opp van sektor $= \frac{1}{2} r^2 \theta$ $= \frac{1}{2} \times \left(\frac{25}{2}\right)^2 \times (1,2)$ $= 93,75 \text{ m}^2$ OR / OF Area of sector / Opp van sektor $= \frac{rs}{2}$ $= \frac{\left(\frac{25}{2}\right) \times 15}{2}$ $= 93,75 \text{ m}^2$	✓ F A ✓ SF CA ✓ Area / Opp CA NPU, NPR AO Full Marks / Volpunte OR / OF ✓ F A ✓ SF CA ✓ Area / Opp CA NPU, NPR AO Full Marks / Volpunte	(3)
9.3.1	$\frac{1200}{1\text{min}} \times \frac{1}{60\text{s}} = 20 \text{ rps}$	✓ Ans / Ant A NPU	(1)
9.3.2	$\omega = 2\pi n$ $\omega = 2 \times \pi \times 20$ $\omega = 40\pi \text{ rad} / \text{s}$ $\omega \approx 125,66 \text{ rad} / \text{s}$	✓ F A ✓ SF CA ✓ Ans / Ant CA	(3)

9.3.3	$v = \pi Dn$ $v = \pi \times \left(\frac{50}{1000}\right) \times 20$ $v = \pi \text{ m/s}$ $v \approx 3.14 \text{ m/s}$	✓F A ✓ SF CA ✓ Conv / Herlei A ✓ Ans / Ant CA	(4)
9.3.4	1 min/ 1 200 rev $\therefore \text{number of revs} / \text{aantal rev} = 30 \times 1200$ $= 36\,000 \text{ revs}$	✓M A ✓ S CA ✓ Ans / Ant CA AO Full Marks / Volpunte	(3)
			[20]

QUESTION/VRAAG 10			
10.1	$A_T = a \left(\frac{o_1 + o_n}{2} + o_2 + o_3 + o_4 + o_5 + o_{n-1} \right)$ $306,60 = \frac{42}{6} \left(\frac{8 + 5}{2} + 7 + 6 + y + 8,7 + 9 \right)$ $306,60 = 7(37,2 + y)$ $\therefore y = \frac{306,60}{7} - 37,2$ $y = 6,6 \text{ m}$ OR / OF $A_T = a(m_1 + m_2 + m_3 + m_4 + \dots + m_{n-1})$ $306,60 = \frac{42}{6} \left(\frac{8+7}{2} + \frac{7+6}{2} + \frac{6+y}{2} + \frac{y+8,7}{2} + \frac{8,7+9}{2} + \frac{9+5}{2} \right)$ $\frac{306,6}{7} = \frac{597}{20} + \frac{14,7 + 2y}{2}$ $\frac{279}{20} = \frac{147}{20} + y$ $\therefore y = 6,6 \text{ m}$	✓F A ✓ SF A ✓ $a = \frac{42}{6}$ A ✓S CA ✓ Area / Opp CA OR / OF ✓F A ✓ SF A ✓ $a = \frac{42}{6}$ A ✓S CA ✓ Area / Opp CA	(5)

10.2				
10.2.1	$(l)^2 = (5,7)^2 + (2)^2$ $l = \sqrt{36,49}$ $l = 6,04 \text{ cm}$	✓ M ✓ Ans / Ant NPR, NPU	A CA	(2)
10.2.2	$A_T = \frac{1}{2}(4\pi r^2) + (\pi r^2 + \pi r l)$ $= \frac{1}{2}(4 \times \pi \times (2)^2) + (\pi \times (2)^2 + \pi \times 2 \times 6)$ $= 24\pi$ $= 75,40 \text{ cm}^2$	✓ F ✓ SF ✓ Ans / Ant	A CA CA	(3)
10.2.3	$V_T = \frac{1}{2}\left(\frac{4}{3}\pi r^3\right) + \frac{1}{3}\pi r^2 h$ $= \frac{1}{2}\left(\frac{4}{3} \times \pi \times (2)^3\right) + \frac{1}{3} \times \pi \times (2)^2 \times 5,7$ $= 40,63 \text{ cm}^3$ $= 40,63 \text{ cm}^3 \times \frac{1l}{1000 \text{ cm}^3}$ $= 0,04 l$	✓ F ✓ SF ✓ Conv/Herlei ✓ Volume in l	A CA CA	(4)
				[14]
			TOTAAL:	150